

WHO AM I?

I am a final-year PhD student at ETH Zurich supervised by Roger Wattenhofer. Interested in human-aligned compression, subjective evaluation of AI-generated images and text, and low-latency machine learning.

EDUCATION

- 2023 - pres. **PhD in Representation Learning** **ETH Zürich**
Image compression, image quality assessment, and multimodal machine learning. Supervising 30 BSc and MSc Thesis. Hands on Deep Learning course coordinator
- 2021 - 2023 **MSc in Mathematical Modelling and Computation** **Technical University of Denmark**
Advanced Machine Learning, Deep Learning, Deep Learning in Image Analysis, Advanced Image Analysis, Bayesian Machine Learning, Signal Processing *GPA:5.73/6*
- 2022 **Study abroad** **ETH Zürich**
Computer Vision Lab: research project in open vocabulary object detection with multi modal machine learning, Seminar in Deep Neural Networks *GPA:5.71/6*
- 2016 - 2020 **BSc in Finance Mathematics and Economics** **Corvinus University of Budapest**
Calculus, Probability theory, Linear Algebra, Statistics, Finance, Modelling, Programming, Microeconomics, Macroeconomics, Game Theory, Negotiation Techniques, etc. *GPA:5.57/6*

EXPERIENCE

- 2026 - 2026 **Research intern** **Mabyduck**
Developing Bayesian models for subjective evaluation data aggregation. Models to detect low-quality human raters. Demonstrating the perception-distortion tradeoff in practice.
- 2020 - 2023 **Computer vision research assistant** **PufinID**
part time Applying contrastive metric learning models for pattern recognition. Building image capturing quality control system. Using latent-diffusion models for synthetic data generation.
- 2019 - 2020 **Quantitative model developer** **Morgan Stanley**
Continuing the internship project, by exploring performance with market frictions, especially looking at model robustness of moving from synthetic training data to real testing data.

SELECTED PUBLICATIONS

- Till Aczel, Lucas Theis, Roger Wattenhofer (2026). "Efficient Bayesian Inference from Noisy Pairwise Comparisons" ICLR 2026
- Lukas Rüttgers, Till Aczel, Andreas Plesner, Roger Wattenhofer (2026). "Light Differentiable Logic Gate Networks" ICLR 2026
- Shakir Yousefi, Andreas Plesner, Till Aczel, Roger Wattenhofer (2025). "Mind the Gap: Removing the Discretization Gap in Differentiable Logic Gate Networks" NeurIPS 2025
- Till Aczel, David Jenny, Simon Bühner, Andreas Plesner, Antonio Di Mario, Roger Wattenhofer (2024). "IC-DLC: Differentiable Logic Circuits for Hardware-Friendly Image Compression" ML4Wireless @ AAI 2026
- Till Aczel, Roger Wattenhofer (2024). "Conditional Hallucinations for Image Compression" Data Compression Conference 2025